Exam 1: Grading/Rubric

ER (15 points)

- Question: The ER data model supports the concept of entities and relationships. What is its formal definition of an entity and a relationship? Provide an example to illustrate these two concepts.
- Answer: Entity is a thing which can be distinctly identified. A relationship is an association / connection among entities.“Father-son” is a relationship between two “person” entities.
- Grader: Xinyan Yu (xinyany@usc.edu), office hour:
  - Tuesday, October 21, **11am - 1pm,** GCS B2 Room **SB6**
  - Wednesday, October 22, **11:30am - 1:30pm**, GCS B2 Room **SB11**
- Rubric:
  - Entity: Mentioning object / thing / stuff / nouns – 5 pts
  - Relationship: Mentioning relates / connects / associates / happens between – 5 pts
    - Mathematical definition – 5 pts
  - Logical representations (table / tuple / attribute sets) – 0 pt
  - Example: reasonable concrete example / drawing out – 5 pts

FoundationDB and Nova-LSM (15 of 30 points)

Question: State two **similarities** between FoundationDB and Nova-LSM.

1. Disaggregated hardware architecture / both use the idea of divide and conquer.
2. Individually scalable components.
3. Key-value data model.
4. Stateless processing / Separate storage from processing.
5. Both range partition data across storage nodes (or if mentioned HRPS).
6. Balance load across the storage nodes by migrating ranges or have load balancing mechanisms.
7. Both use logging for durability, availability, and failure handling.
8. Data replication to ensure availability and durability.
9. (Wrong but awarded points) Both have shared nothing architecture.
    a. advanced topic: Nova-LSM uses RDMA to share memory across the nodes.

Grader: Hamed Alimohammadzadeh, halimoha@usc.edu, Office Hours:

b. Wednesday, October 22, **3:00 – 5:00pm**, GCS B2 Room **SB8**
c. Thursday, October 23, **10:00am – 12:00pm**, GCS B2 Room **SB7**

Rubric: **15 points** for any 2 correct item, **8 points** for any 1 correct item.

FoundationDB and Nova-LSM (15 of 30 points)

Question: State two **differences** between FoundationDB and Nova-LSM.

1. transaction related (any of below items):
   a. FDB supports transactions at the granularity of multiple read and write sets vs transactions are at the granularity of either one read or one write.
   b. FDB provides ACID transactions.
   c. FDB has crash recovery protocol.
   d. FDB has strict serializability using Multi-Version Concurrency Control (MVCC) and Optimistic Concurrency Control (OCC).
2. FDB uses off-the-shelf hardware vs Nova-LSM uses RDMA.
3. FDB spans multiple datacenters vs one data center.
4. Nova-LSM is optimized for write-heavy loads.

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Rubric: **15 points** for any 2 correct item, **8 points** for any 1 correct item.

Gamma and Grasper (20 of 40 points)

- Question: State three similarities between Gamma and Grasper.
- Answer:
  - Both partition data across nodes for load balance.
  - Both have inter-operator **pipelining** (stream) for throughput.
  - Both can be scaled up by **distributing both data and computation**.
  - Both hide **parallelism** using a query language.
  - Both support the **concept of operators** that process data in a pipeline manner.
  - Grasper's **coordinator** is similar to Gamma's **scheduler**.
    - Both coordinate the execution of queries.
  - Both **divided complex** queries into smaller operations (steps).
  - Both Gamma and Grasper **minimize communication**. Gamma sends the query to the node with relevant data, Grasper requires Experts to cache data to minimize communication.
  - Both use **divide and conquer**.
  - **(Wrong but awarded points)** Both use Shared-Nothing Architecture (Grasper uses RDMA to share memory).
- Grader: Shuqin Zhu (shuqinzh@usc.edu)
  - Office hours:
    - Wednesday, October 22, **2:00 – 3:00pm**, GCS B2 Room **SB9**
    - Wednesday, October 22, **3:30 – 4:30pm**, GCS B2 Room **SB5**
    - Friday, October 24, **12:00 – 2:00pm**, GCS B2 Room **SB4**
- Rubrics: **20 points** for any 3 correct items, **14 points** for 2, **8 Points** for 1.

Gamma and Grasper (20 of 40 points)

- Question: State three differences between Gamma and Grasper.
- Answer:
  - Gamma uses a **Shared-Nothing Architecture,** where Grasper uses RDMA to share memory (each node sets aside a portion of its memory to be used by other nodes).
  - Gamma is **relational** DB, Grasper is **graph** DB.
  - Gamma supports **SQL**, Grasper supports **Gremlin**.
  - Gamma is **disk** based, Grasper in an **in-memory** system.
  - Grasper used **RDMA**, Gamma used off the shelf hardware.
  - Gamma uses a **query tree**, Grasper uses **query DAG**.
  - Gamma's **parallelism is fixed**, parallelism for Grasper **adepts per step**.
  - Grasper has a **master** for load balancing, Gamma does not have this concept.
  - Grasper uses **works stealing** to balance the load, Gamma does not have such mechanism
  - Grasper **caches data** from other nodes, Gamma ships a query to the node with the **relevant data**.
  - Gamma is built for **read and write**, but Grasper is **read only**.
  - Gamma enhances **failure management** through **chained declustering**, where each fragment's **backup** is stored on a different node. Grasper does not have failure recovery.
  - Gamma supports **transactions**, Grasper does not.
  - Gamma partitions data across nodes using **hash, range, or round-robin methods**, Grasper uses hash only.
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- Rubrics: **20 points** for any 3 correct items, **14 points** for 2, **8 Points** for 1.

HRPS (15 points)

- Question: Consider a very selective query that fetches one record from a very large table (tens of Petabytes in size) using an index structure, say a B+-tree index. The table is partitioned across 200 nodes in a data center. What happens to the query response time and system throughput if this query is directed to all 200 nodes for processing? Make sure to explain your answer.
- Answer: Query response time increases, system throughput decreases. Coordinating 200 nodes increases response time. The throughput decreases because 199 nodes search their fragment to find no qualifying record.
- Grader: Wen Yan (ywen8094@usc.edu), office hour:
  - Friday, October 24, **9am - 11am,** GCS B2 Room **SB11**
- Rubric:
  - Query response time increases / worsens – 2.5 pts
  - System throughput decreases / worsens – 2.5 pts
  - Mentioning **overhead** / wasted resource / waiting for unnecessary nodes / takes longer to access data – 5 pts
  - Mentioning only **one** / minimal number of **nodes** needs to be hit / **199 nodes** doing nothing – 5 pts
    - (repeating the question) one record / selective query without mentioning one **node** – 0 pts

We tried our best to give partial credits